

The Intention Superiority Effect in Motor Skill Learning

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Three experiments were conducted to determine if the intention to perform motor sequences in the future results in similar patterns of activation and inhibition as observed for verbal scripts. In Experiments 1 and 2, intention was induced by informing one group that they would be tested on the tasks following acquisition; the other group was not informed of the retention test. Recognition tests administered prior to and after the retention test indicated a strong intention superiority effect. However, intention instructions provided either at the end of acquisition (Experiment 1) or before acquisition (Experiment 2) failed to impact acquisition or retention performance of the motor sequences, but did influence the latency of responding on the retention test. Experiment 3 was designed to replicate the results of Experiments 1 and 2 using a within-subjects design and extend these findings to observation. The results indicated that intention instructions resulted in a strong *intention superiority effect* for both the physical and observational practice participants, but the performance on the intentional tasks was enhanced only for the observational practice group.

Keywords: intention superiority effect, prospective memory, retrospective memory, motor skill learning

The term *intention* has been used in the verbal literature to denote that an individual has developed a need or desire to perform an activity at some time in the impending future (e.g., Goschke & Kuhl, 1993; Marsh et al., 1999). Intentions are often referred to as prospective memories because the memory is stored and encoded with the intention to be used some specific time or place in the future. Indeed, intention research using prospective memory tasks often classifies the tasks as event-based prospective tasks when the intent to execute the task is dependent on some future event and as time-based prospective tasks when the intent is dependent on some future time (Einstein & McDaniel, 1990). Examples of event-based intentions, in this context, may include individuals' needs to remember to take their medication after dinner, pick up a package on the way home from work, or give a message to a colleague the next time they run into him or her. Examples of time-based intentions include meeting a business associate for dinner at 7:00 p.m. or turning the lawn sprinklers off after 20 minutes. Prospective memory is contrasted with retrospective memory, which is memory requiring the recall or recognition of past information, events, and so forth. Retrospective and prospective memories appear to be supported by separate neuroanatomical pathways (Cockburn, 1995; Shimamura, 1994).

Although relatively little is known about how intentions (prospective memories) are represented and stored in memory, recent research on the activation levels of delayed and fulfilled intentions to execute verbal scripts has provided some important clues. More specifically, a number of experiments have demonstrated that

delayed intentions hold a privileged place in memory such that the intention to perform a verbal script, for example, results in the script being held in a heightened state of activation prior to completion as indicated by a recognition test (e.g., Goschke & Kuhl, 1993) or lexical-decision task (e.g., Marsh et al., 1999). After the intention has been fulfilled or cancelled, it appears inhibited, at least relative to more neutral material (e.g., Dockree & Ellis, 2001; Marsh et al., 1999). This effect has been termed the *intention superiority effect*.

Goschke and Kuhl (1993), for example, asked participants to learn two of four procedural scripts, each consisting of five specific steps to complete a simple activity (e.g., steps to set the dining table or clear a messy desk). After learning the script, participants were informed that they would have to perform one of the scripts in a later test. These instructions were designed to induce intention for the script to be performed in the future, while the other script remained neutral with respect to intention. In a recognition test administered subsequent to the intention instructions, participants were presented key words from the two scripts and semantically similar words not related to the scripts. The task was to respond as quickly as possible using one response key if the words were contained in the learned scripts and another response key if they were not. The important comparison, however, was between the latencies for the words taken from the intentional (to-be-performed) script versus the latencies for the words taken from the neutral script. The results indicated significantly shorter latencies on the recognition test to the words in the to-be-performed scripts than to the words in the other learned, but neutral, lists. Upon completion of the recognition test, a retention test was administered in which the participants recalled the scripts.

In additional experiments, Goschke and Kuhl (1993) demonstrated that the intention superiority effect persisted even when distractor tasks (counting backward in steps of 3 for 45 s) were used to block the selective encoding, post-study imagery, and

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additional rehearsal of the to-be-performed scripts. The effect also persisted when test expectancy was controlled by informing participants in advance that a free recall test would be administered for both scripts. However, the performance of observers in a control condition failed to demonstrate latency differences between the to-be-performed and neutral scripts. Although it is very difficult to rule out all forms of selective encoding of the prospective items, these findings are consistent with the notion that the intention to perform a script in the future results in heightened levels of subthreshold activation and are not the result of selective encoding strategies or additional processing of the to-be-performed task.

Marsh, Hicks, and Bink (1998) replicated the findings of Goschke and Kuhl (1993) in terms of heightened activation levels associated with intention instructions relative to the activation levels associated with more neutral scripts. However, they administered a second lexical decision test after the intention had been fulfilled or cancelled. Their results indicated that after the intention was satisfied the memory became inhibited relative to more neutral material not associated with any specific intention. Marsh et al. (1999) also found inhibition when the intention was cancelled. They suggested that it is adaptive for uncompleted intentions to remain in a heightened state of activation until completion, after which the memories underpinning the task are inhibited. In this way, other unfulfilled intentions would come to mind more readily than intentions already fulfilled.

An additional, but perhaps unrelated, effect of intention involves the effect of intention on the actual performance of the intended task rather than on just the recognition of the task. In most, if not all, of the intention experiments we are aware of in the verbal literature, the intention was induced after the initial encoding of the task (after practice but before the delayed production of the task). Presumably, this design feature was used so as to not differentially influence the later performance of the task. Enhanced performance of the intended scripts relative to the neutral scripts on the delayed test could potentially confound the interpretation of the recognition latencies. However, it is possible that the encoding processing associated with the learning of motor tasks continues for some time after the completion of practice (e.g., Karni, Tanne, Rubenstein, Askenasy, & Sagi, 1994; Tully, Preat, Boynton, & Del Vecchio, 1994) such that the inducement of an intention after practice may result in additional processing of the intended tasks, deficient processing of the tasks that are neutral with respect to intention, or both. Of course, if intentions are formed during practice (during encoding), the manner in which the prospective and neutral tasks are processed is more likely to be different.

In addition, it is important to determine if intentions affect learning by observation in the same manner that they affect learning during physical practice. Recent neuroimaging experiments report that a set of common neural structures is activated during both action production and action observation (Gallese & Goldman, 1998; Grèzes & Decety, 2001; see Jeannerod, 1999, for reviews). The shared neural structures included the premotor cortex, supplementary motor area, inferior parietal lobule, cingulate gyrus, and cerebellum. This finding has led Jeannerod (1999, 2001) to suggest a degree of functional equivalence between action generation, action simulation, action verbalization, and perception of action. For example, Jeannerod (1999) proposed that "Activation of motor structures might therefore represent a plausible

neural basis for motor representations subserving, not only motor imagery, but also imitation and observational learning" (p. 10). The activation of cognitive and motor neural structures, in the absence of overt action, provides a possible explanation for the beneficial effects of observation on motor learning. According to Bandura (1986), observation of a model's performance enables the observers to acquire a representation of the task to be learned. Afterward, when the observers are required to perform the motor task physically, this representation is used to select and to program the required response. The movement representation and associated processing mechanisms acquired via observation are thought to be similar to those developed during physical practice (Schmidt, 1975; Schmidt & Lee, 1999). On this basis, it was suggested that observational learning and learning through physical practice may be mediated by similar cognitive processes (Adams, 1986). This notion is consistent with results that have shown that variables affecting learning through physical practice tend to affect observational learning in a similar way. For example, the schedule of practice experienced during observation (Blandin, Proteau, & Alain, 1994; Wright, Li, & Coady, 1997), the sensory information available during observation (Shea, Wulf, Park, & Gaunt, 2001), and the schedule by which knowledge of results (error information) was presented (Badets & Blandin, 2004, 2005) have produced similar patterns of results in learning for models and observers. Therefore, it was suggested that both observation participants and physical practice participants engage in similar cognitive processes responsible for action planning and control. Thus, we are interested in determining if the performance manifestations of the intention superiority effect are similar for physical and observational practice.

Understanding the boundary conditions for the effect and the mechanisms involved in maintaining and executing delayed intentions is of practical importance because individuals in our society need to remember important intentions to function effectively. A clear theoretical picture is also important to the understanding of the decline in the ability to remember intentions that sometimes occurs in the course of aging (e.g., Kidder, Park, Hertzog, & Morrell, 1997; Rendell & Craik, 2000) or dementia and is associated with specific psychological problems (e.g., Kondel, 2002; Maylor, Darby, & Della Sala, 2000). Clearly, severe deficits in one's ability to remember delayed intentions can jeopardize an individual's ability to live independently. For example, forgetting to turn the stove off when heating water for tea or forgetting that the bath water was running could have severe consequences. Understanding how intentions are maintained and appropriately acted upon is also important from a theoretical perspective because disorders affecting the maintenance of intention will remain difficult to diagnose and effectively treat until the underlying mechanisms are reasonably well understood. In addition, it is important to determine if intention superiority effects are a general phenomenon that impacts other procedural tasks including the intention to execute a motor sequence in the impending future or whether the effects are restricted to specific types of verbal tasks like the ones used in previous experiments (e.g., Goschke & Kuhl, 1993; Marsh et al., 1999).

Thus, the primary purpose of the following experiments was to determine if an intention superiority effect would be found for the learning of relatively simple timing sequences. Participants were provided practice on four simple timing sequences, each composed

of five elements. The timing task used in the present experiments differed from the procedural tasks used in many of the previous experiments in a number of ways. First, in contrast to many of the verbal scripts used in previous experiments in this literature, the timing tasks are probably relatively novel to the participants. Thus, participants use the practice session to develop a memory for the timing sequence. This is in contrast to many of the verbal scripts for procedures like setting a table or tidying a messy desk, for example, which were probably relatively familiar to the participants. Second, the timing tasks are encoded motorically rather than verbally. Indeed, the timing tasks do not readily lend themselves to verbal description, although this is not impossible. We have used these tasks before, and participants when asked to describe the tasks in postexperiment interviews most often tap the sequence with their finger or repeat a simple sound (like “da”) with specific temporal intervals between the taps or sounds (e.g., Shea et al., 2001). It seems like verbal description of the tasks, while possible to construct, are not directly related to the way in which participants code the sequences in memory. Finally, a quantitative analysis of the timing tasks was conducted. A quantitative analysis as opposed to a qualitative analysis is important in order to determine if intention instructions have subtle impacts on performance of the task or whether the effect is restricted to the latencies observed on the recognition tests administered before and after the retention test. In addition, in the present experiments substantially more practice is provided on the tasks than in many of the previous experiments.

After the completion of the acquisition phase (Experiment 1) or before the acquisition phase (Experiment 2), one group of participants was informed that they would be required to perform the sequences at a later time in a retention test. Another group of participants was not informed about the retention test. The information about the retention test was designed to induce intention for the sequences, while the sequences, presumably, remained neutral with respect to intention for the group not provided this information. The location of the intention instructions was manipulated because we hypothesized that providing the instructions at the end of practice would not impact the way in which participants encoded task information and therefore not affect performance of the tasks. Providing this information before practice, however, may change the way in which the participants process and store task-relevant information—leading to performance differences between the intention and no-intention groups. We initially formed this hypothesis on the basis of a recent experiment of ours (Badets, Blandin, & Shea, in press), which demonstrated that the intention to perform a movement sequence in the future enhanced the observational learning of motor skills. Presumably, physical practice would be similarly impacted. In Experiment 3, we attempted to replicate the findings from Experiment 2 using a within-subject design in which intention was induced for two sequences, but not for two others that were practiced during acquisition. In addition, an observer was yoked to each physical practice participant to determine if intention affects physical and observational learning similarly. Note also that these experiments utilize between-subjects (Experiments 1 and 2) and within-subjects (Experiment 3) designs. The different designs were used to ensure that the effects were not design specific. Jahn and Engelkamp (2003) have noted discrepancies in prospective memory experiments related to within- and between-subjects designs.

Experiment 1

The intention superiority effect has been demonstrated using only a limited number of prospective tasks and paradigms. If this is a general memory phenomenon, one would expect to see similar patterns of activation and inhibition regardless of the task and paradigm. Thus, our primary goal in this experiment was to determine if the memories underpinning a learned movement sequence, which is intended to be performed in the impending future, will be stored in a heightened state of activation prior to the fulfillment of the intention to perform the task and then become inhibited, relative to more neutral tasks, after completion of the intent. We also wanted to determine if intent affected in any way the performance of the task or influenced the time required to retrieve, process, or otherwise program the movement sequence. Presumably, the activation level of specific memories will be observed not just on recognition tests, but also in the time required to ready a movement sequence for execution.

To test these hypotheses, we used a between-subjects design. In most of the verbal script learning experiments, a within-subjects design was used such that the relative activation levels of the proscriptive scripts were compared to those of the more neutral scripts for the same participant. In this experiment, participants were randomly assigned to either an intention group or a no-intention group. In this design, everything, including the proscriptive tasks and associated items used in the recognition tests, was identical except the instructions used to induce intention. We are not proposing that this design was superior to that used in previous experiments, only that important cognitive-motor phenomenon should be shown to be paradigm independent.

Method

Participants. University students ($N = 18$) volunteered to participate in the experiment (mean age 21.5 years, $SD = 1.1$ year). None of the participants had prior experience with the task or was informed about the purpose of the experiment. Each participant was requested to read and sign an informed consent form prior to participation in the experiment.

Apparatus and task. The apparatus consisted of a Pentium IV computer with a 21-in color monitor and a standard keyboard. The task required the use of the *F* and *J* keys on the keyboard with the first finger of the left and right hand, respectively. The computer monitor was used to display the goal movement pattern (Figure 1A) and to provide feedback to the participant (1B). A customized computer program controlled all the experimental procedures, provided feedback to the participants, and stored the data for further analysis. The task was to alternately press the keys in an attempt to produce as closely as possible the goal pattern displayed on the screen. The goal pattern required six keypresses to produce the 5-element sequence. Each element was defined as the time from one keypress to the next. Four sequences were practiced, which were differentiated in terms of the hand that initiated the sequence (left or right) and the duration of the response (long or short). Thus, the practice sequences were named long-left-1, short-left-1, long-right-1, and short-right-1. The practice sequence patterns are displayed in Figure 1C. For the long-duration tasks (1600 ms) the goal times between consecutive keypresses were 300, 500, 200, 200, and 400 ms, and for the short-duration tasks (800 ms) the goal times between keypresses were 150, 250, 100, 100, and 200 ms.

Four distractor sequences were constructed from the same elements used to construct the practice sequences. These sequences were used only as distractors in the recognition tests and were not practiced. The distractor sequences were named long-left-2, short-left-2, long-right-2, and short-right-2. The distractor sequence patterns are displayed in Figure 1D. For

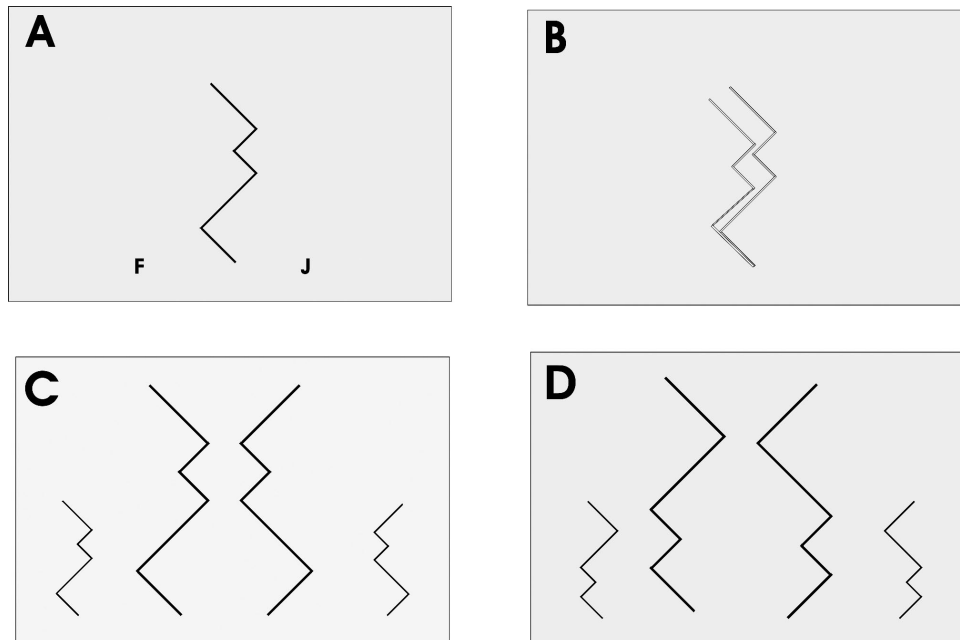


Figure 1. Illustration of the screen display and sequence patterns used in all experiments. Panel A illustrates the screen display prior to the participant's response, and Panel B illustrates the screen display after the feedback is provided. Note that in Panel B Element 1 was produced too slowly by the participant and Element 2 was produced too quickly. Elements 3–5 were produced correctly. Panel C illustrates the practiced sequence patterns used during acquisition, retention, and in the both recognition tests. Panel D illustrates the distractor patterns used in both recognition tests.

the long-duration distractor tasks (1600 ms) the goal times between consecutive keypresses were 300, 200, 200, 500, and 400 ms, and for the short-duration distractor tasks (800 ms) the goal times between keypresses were 150, 100, 100, 250, and 200 ms.

Procedure. Participants were randomly assigned to one of two groups ($n = 9$ for each group). The groups differed only in terms of the intention instructions provided at the end of acquisition. Thus, the group that received the intention instructions was labeled the intention group and the other was labeled the no-intention group. Prior to testing, all participants received the same verbal instructions regarding the goals of the task for the acquisition and retention phases of the experiment.

Participants were informed that pressing the *J* key would cause the line to move to the left in an upward direction at a constant velocity, while pressing the *F* key would cause the line to move to the right in an upward direction at a constant velocity. Participants were told that they could begin their response anytime after the goal pattern was displayed (see Figure 1A). The goal pattern remained on the monitor during the response and for 8 s after the completion of the response. During the response, the line produced by the participant was not displayed. However, upon the completion of each acquisition response, the pattern produced by the participant was superimposed on the target pattern for 5 s. This feedback provided information concerning timing errors (see Figure 1B). Prior to the first trial, the experimenter explained the feedback that would be provided using illustrations of goal patterns with hypothetical feedback overlaid. Participants were told to attempt to match the pattern as closely as possible.

Before beginning the acquisition phase, all participants were informed that they would be required to complete two recognition tests (reaction time tasks) approximately 10 minutes after completion of the practice session. All participants performed 80 practice trials during the acquisition phase (20 trials per task in a random order). After completing the acquisition session, participants in the intention group were informed that they

would be required to reproduce the practice tasks at the end of the first recognition test. The specific instruction was “in 10 minutes, you will have to reproduce the tasks which you have just practiced.” This last instruction was not provided to the participants in the no-intention group. At the end of the retention interval, participants performed a recognition test consisting of 40 practice tasks (10 for each task) and 40 distractor tasks (10 of each). The practiced and distractor tasks were presented in a random order. On each trial in the recognition test, a goal movement pattern was displayed. The participant's task was to depress one key if the task was one that had been practiced and another if it had not been practiced. The practiced tasks were considered prospective tasks for the intention group and neutral with respect to intention for the no-intention group. The distractor tasks were considered distractors for both groups. The practiced and distractor patterns used in the recognition test are displayed in Figure 1C and D, respectively. After this recognition test, all participants were asked to complete a retention test consisting of 40 trials. The retention test was conducted in a manner identical to that of acquisition. After the retention phase, all participants performed the second recognition test in the same manner as the first.

Dependent variables. For acquisition and retention, the dependent variable of primary interest was timing error. Timing error was computed as the average of the absolute differences between the goal element durations (GED) and the actual element durations (AED). This measure provides an estimate of the accuracy with which participants produced the goal timing pattern. In equation form,

$$\text{Timing error} = (|AED_1 - GED_1| + |AED_2 - GED_2| + |AED_3 - GED_3| + |AED_4 - GED_4| + |AED_5 - GED_5|)/5$$

where AED_n is the actual element duration, and GED_n is the goal element duration for Elements 1–5. In addition, the time from the presentation of

the goal pattern to the depression of the first key was recorded on the retention test. This measure, termed T1, indicated the time required to retrieve, prepare, or otherwise program the response.

For the recognition tests, reaction time, the time from the presentation of the goal pattern until the participant responded, was the primary dependent variable. In addition, percentage correct, the number of times the participant responded correctly (correct rejections of distractor patterns and hits for the practiced patterns), was computed and analyzed. A yes-no d prime and criterion c (a measure of bias) were also computed and analyzed.

Results

Acquisition and retention. As anticipated, there did not appear to be any differences between groups during acquisition or retention (Figure 2A and C, respectively). Timing errors during acquisition were analyzed in a 2 Group (intention vs. no intention) $\times$ 4 Block (1–4) analysis of variance (ANOVA) with repeated measures on block. The analysis failed to detect a difference between groups, $F(1, 16) < 1$, $p > .05$, or a Block $\times$ Group interaction, $F(3, 48) < 1$, $p > .05$. The main effect of block was significant, $F(3, 48) = 17.66$, $p < .01$. Duncan's new multiple-range tests indicated that timing errors were higher in Block 1 than in Block 2, which were both higher than in Blocks 3 and 4, which did not differ from each other. Timing errors and T1 (the time to initiate the first keypress) during retention were analyzed in Group (intention vs. no intention) $\times$ Block (1–2) ANOVAs with repeated measures on block. Timing errors on the retention test also did not differ across groups, $F(1, 16) < 1$, $p > .05$, or block, $F(1, 16) = 1.49$, $p > .05$. The Group $\times$ Block interaction also was not significant, $F(1, 16) = 3.71$, $p > .05$. However, the analysis of T1 indicated that participants in the intention group, $F(1, 16) = 17.16$, $p < .01$, initiated their responses more quickly than participants in the no-intention group (Figure 2F). The main effect of block, $F(1, 16) = 3.11$, $p > .05$, and Group $\times$ Block interaction, $F(1, 16) = 3.03$, $p > .05$, for T1 was not significant.

Recognition tests. Recognition test reaction times (Figure 2B and D) and percentage correct (Figure 2E and G) were analyzed in a 2-Group (intention vs. no intention) $\times$ 2-Task (practiced vs. distractors) $\times$ 2-Test (preretention vs. postretention) ANOVA with repeated measures on task and test. The analysis of reaction time failed to detect the main effects of group, $F(1, 16) = 1.60$, $p < .05$, or task, $F(1, 16) < 1$, $p > .05$. The main effect of test was significant, $F(1, 16) = 33.37$, $p < .01$, with the reaction times substantially shorter on the postretention recognition test than on the preretention recognition test. While the Group $\times$ Task, $F(1, 16) < 1$, $p > .05$, and the Task $\times$ Test, $F(1, 16) < 1$, $p > .05$, interactions were not significant, the analysis detected a Group $\times$ Task $\times$ Test interaction, $F(1, 16) = 4.60$, $p < .05$.

Simple main effect analysis on the preretention recognition test failed to detect differences between distractor and practiced tasks for the no-intention group but did detect shorter reaction times for the practiced (prospective) tasks than the distractors for the group provided intention instructions at the end of acquisition. On the postretention recognition test, simple main effect analyses again failed to detect differences between distractors and practiced tasks for the no-intention group but did detect differences for the intention group.

The analysis of percentage correct failed to detect a main effect of group or task. However, the main effect of test, $F(1, 16) = 4.73$, $p < .05$, was significant with percentage correct slightly lower on

Recognition Test 1 than on Recognition Test 2 (Figure 2E and G). All interactions failed significance. The average d prime on Recognition Test 1 for the no-intention group was 1.79, and the d prime for the intention group was 1.95. On Recognition Test 2, average d prime for the no-intention group was 2.48, and the d prime for the intention group was 3.50. d prime and Criterion C were analyzed in a Group $\times$ Test ANOVA with repeated measures on test. For d prime, the main effects of group, $F(1, 16) = 4.09$, $p > .05$, and the Group $\times$ Test interaction, $F(1, 16) = 2.57$, $p > .05$, were not significant. However, a main effect of test was detected, $F(1, 16) = 17.82$, $p < .01$, indicating an increase in d prime from Recognition Test 1 to Recognition Test 2. No significant effects were detected in the analysis of Criterion C.

Discussion

The primary purpose of the experiment was to determine if prospective motor tasks are stored in a heightened state of activation before the intention is fulfilled and then are inhibited relative to distractor tasks. A secondary purpose was to determine if intention instructions influenced in any way the performance of the prospective tasks.

The pattern of results on the recognition tests administered prior to and after the retention test was consistent with the intention superiority effect found for verbal scripts (e.g., Goschke & Kuhl, 1993; Marsh et al., 1998). That is, recognition of the practiced (prospective) tasks prior to the retention test resulted in latencies that were significantly shorter than the latencies for the distractor tasks for the intention group. No differences were detected between practiced and distractor task latencies for participants not provided intention instructions. Indeed, the latencies to the practiced tasks were significantly shorter for the intention group compared to the no-intention group. Further, the performance of the intention group on the postretention recognition test indicated longer latencies for the practiced (prospective) tasks than for the distractor tasks. Again, this pattern was not observed for the no-intention group. Latencies for the practiced tasks, however, did not differ across groups. This pattern of heightened activation prior to fulfillment of an intention and inhibition after fulfillment is the hallmark of the intention superiority effect. It is important to note that the recognition test results were not confounded by differences in percentage correct. That is, it could be possible for this pattern to arise as a result of participants trading off (in order to respond more quickly) recognition accuracy for speed of responding. If participants had responded less accurately to the prospective tasks than to the distractor tasks, then the finding of decreased latencies would be suspect. However, this was clearly not the case. The percentage of correct responses on both recognition tests was virtually identical for the prospective and distractor tasks. Thus, it appears that the intention superiority effect may be a general memory phenomenon and not restricted to specific types of procedural tasks.

Performance on the tasks during acquisition was not affected by the intention instructions provided after acquisition was completed. We speculated that providing these instructions might have resulted in additional processing of the tasks to be performed in the future. However, the retention analysis failed to detect any performance differences between participants who were provided the intention instruction and those who were not provided these in-

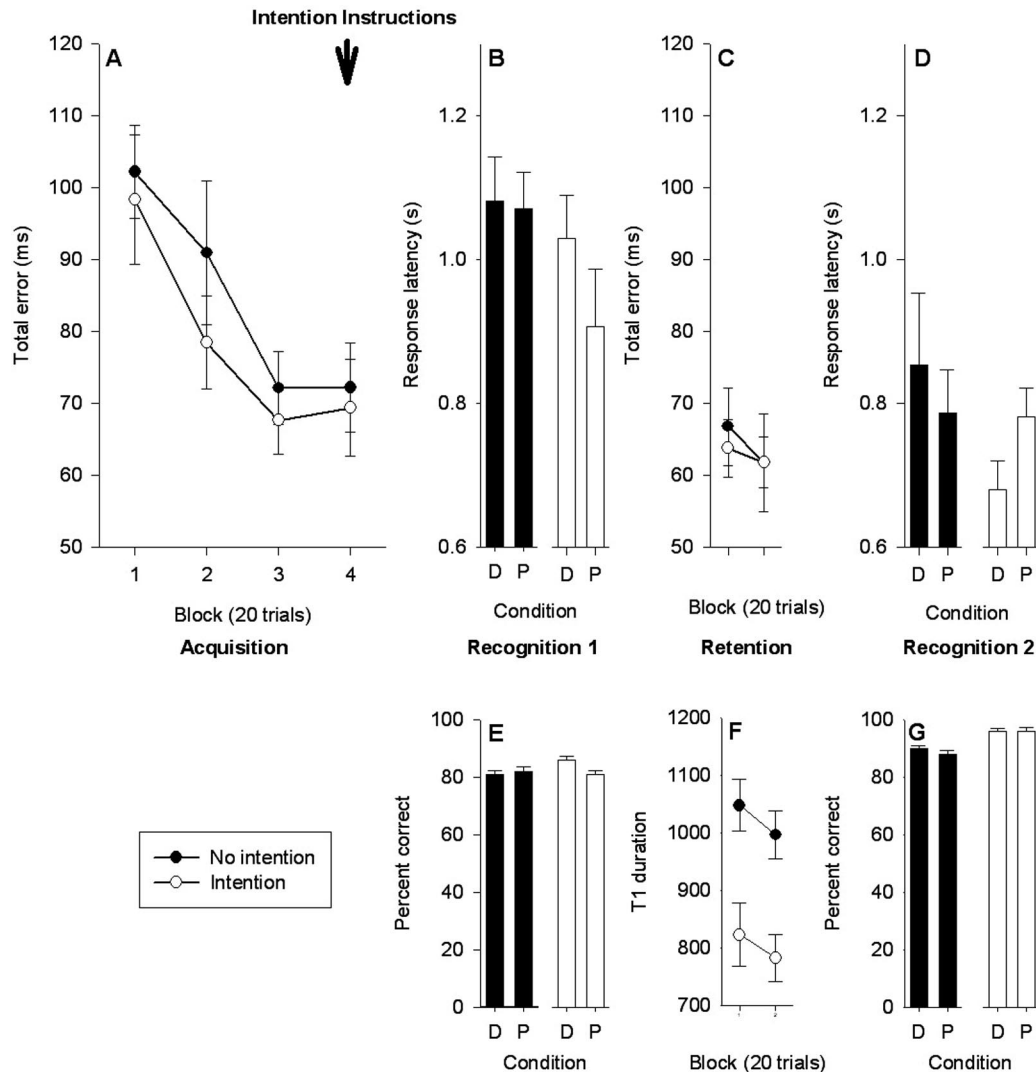


Figure 2. Mean total error during acquisition and retention for Experiment 1 are illustrated in Panels A and C, respectively. Mean response latency on Recognition Tests 1 and 2 are illustrated in Panels B and D, respectively (D indicates distractor patterns, and P indicates practiced patterns). Mean percentage correct on Recognition Tests 1 and 2 are illustrated in Panels E and G, respectively. Time required to initiate responses (T1) on the retention test is provided in Panel F. Intention instructions were provided to the intention group at the end of acquisition. Error bars represent standard errors.

structions. One interesting performance difference between the two groups was noted. Participants were asked to respond as soon as possible after the display (indicating which response to produce) was presented. It is interesting that participants who were provided intention instructions initiated their responses a little more than 200 ms faster on the retention test than the participants who were not provided intention instructions. This is consistent with the findings of decreased latencies for the prospective tasks on the first recognition test. Thus, it appears that intention affects the activation and inhibition of the memories underpinning the tasks and the time required to prepare a response for execution, but not the memory states which underpin response production. In other words, the intention affects the recognition, retrieval, and initial processing of the to-be-performed task, but not the execution of the

task per se. This finding is consistent with the notion that usually, when people establish an intention to perform some action in the future, prospective remembering will concern the conditions under which this intention can be fulfilled (a specific time or event), but not the specific way to fulfill the intention (Kvavilashvili, 1992).

Experiment 2

The data from Experiment 1 revealed patterns of activation consistent with the intention superiority effect when the intention instructions were provided after initial encoding (at the end of practice). This is the position in the experimental sequence where intention had typically been induced in the verbal script learning experiments (e.g., Goschke & Kuhl, 1993; Marsh et al., 1998). The

results from verbal script learning experiments and the present experiment were that performance, per se, was not affected by intention instructions. The primary purpose of Experiment 2 was to determine if providing these instructions before practice might impact not only the activation levels, but also task performance. That is, would knowledge that the task would have to be performed in the future impact the performance and learning of the task? We have observed that providing this type of instruction before observational learning enhanced the observers' performances when they were later required to perform the tasks physically (Badets et al., in press), but to our knowledge effects of intention before encoding on later activation patterns have not been directly tested. There is, however, evidence in the motor learning literature that informing participants that they will be tested on the practiced tasks influences the processing of the task during acquisition (Morgan, 1981). The result was generally better performance on the retention test (no effect on acquisition performance) for the group informed that they would be tested relative to the group that was not provided this information (see Shea & Zimny, 1983, for a discussion). Consistent with these findings, we hypothesized that participants provided intention instructions prior to acquisition may encode task information differently, perhaps more effectively or efficiently, than when intention is not present. The result of additional or more effective processing of task information during acquisition would be enhanced performance during acquisition and enhanced learning as indicated by performance on the retention test.

Method

Participants. University students ($N = 18$) volunteered to participate in the experiment (mean age 21.4 years, $SD = 1.2$ years). None of the participants had prior experience with the task or was informed about the purpose of the experiment. Each participant was requested to read and sign an informed consent form prior to participation in the experiment.

Apparatus and task. The apparatus and tasks were identical to those used in Experiment 1.

Procedure. As in Experiment 1, participants were randomly assigned to one of two groups ($n = 9$ for each group). The groups differed only in terms of the intention instructions provided. However, in Experiment 2 the intention instructions were provided prior to the practice session rather than at the end of practice as in Experiment 1. In all other respects, the procedures were identical.

Dependent variables. The dependent variables were identical to those used in Experiment 1.

Results

Acquisition and retention. We had anticipated that providing intention instructions prior to practice would result in improved performance during acquisition and retention; however, there did not appear to be any differences between the intention and no-intention groups (Figure 3A and C, respectively). Timing errors during acquisition were analyzed in a 2-Group (intention vs. no intention) $\times$ 4-Block (1–4) ANOVA with repeated measures on block. The analysis failed to detect a difference between groups, $F(1, 16) < 1$, $p > .05$, or a Block $\times$ Group interaction, $F(3, 48) < 1$, $p > .05$. The main effect of block was significant, $F(3, 48) = 40.00$, $p < .01$. Duncan's new multiple-range tests indicated that errors were higher in Block 1 than in Block 2, which were both

higher than in Blocks 3 and 4, which did not differ from each other. Timing errors and T1 during retention were analyzed in Group (intention vs. no intention) $\times$ Block ANOVAs with repeated measures on block. Timing errors on the retention test also did not differ across groups, $F(1, 16) < 1.81$, $p > .05$, or blocks, $F(1, 16) < 1$, $p > .05$. The Group $\times$ Block interaction was also not significant, $F(1, 16) < 1$, $p > .05$. The analysis of T1 (Figure 3F), however, indicated, $F(1, 16) = 15.68$, $p < .01$, that participants in the intention group initiated their responses on the retention test sooner than participants in the no-intention group. The main effects of block, $F(1, 16) = 3.01$, $p > .05$, and Group $\times$ Block, $F(1, 16) = 2.15$, $p > .05$ interaction were not significant.

Recognition tests. Recognition test reaction times (Figure 3B and D) and percentage correct (Figure 3E and G) were analyzed in 2-Group (intention vs. no intention) $\times$ 2-Task (practiced vs. distractors) $\times$ 2-Test (preretention vs. postretention) ANOVAs with repeated measures on task and test. The analysis of reaction time failed to detect the main effects of group, $F(1, 16) < 1$, $p < .05$, or task, $F(1, 16) = 1.02$, $p > .05$. The main effect of test was significant, $F(1, 16) = 21.97$, $p < .01$, with the reaction times substantially shorter on the postretention recognition test than on the preretention recognition test. While the Group $\times$ Task, $F(1, 16) = 1.30$, $p > .05$, Group $\times$ Test, $F(1, 16) < 1$, $p > .05$, and Task $\times$ Test, $F(1, 16) = 1.57$, $p > .05$, interactions were not significant, as in Experiment 1 the analysis detected a Group $\times$ Task $\times$ Test interaction, $F(1, 16) = 4.97$, $p < .05$.

Simple main effect analysis on the preretention recognition test failed to detect differences between distractors and practiced tasks for the no-intention group but did detect shorter reaction times for the practiced (prospective) tasks than the distractors for the group that was provided intention instructions at the end of acquisition. On the postretention recognition test, simple main effect analyses again failed to detect differences between distractors and practiced tasks for the control group but did detect differences for the intention group.

The analysis of percentage correct failed to detect main effects of group or task. As in Experiment 1, however, the main effect of test, $F(1, 16) = 4.77$, $p < .05$, was significant with percentage correct lower for both groups on Recognition Test 1 than on Recognition Test 2 (Figure 3E and G). None of the interactions reached significance. The average d prime on Recognition Test 1 for the no-intention group was 1.83, and the d prime for the intention group was 1.79. On Recognition Test 2, average d prime for the no-intention group was 2.56, and the d prime for the intention group was 3.03. d prime and Criterion C were analyzed in a Group $\times$ Test ANOVA with repeated measures on test. For d prime the main effects of group, $F(1, 16) = 0.81$, $p > .05$, and the Group $\times$ Test interaction, $F(1, 16) = 1.05$, $p > .05$, were not significant. However, a main effect of test was detected, $F(1, 16) = 16.39$, $p < .01$, indicating an increase in d prime from Recognition Test 1 to Recognition Test 2. No significant effects were detected in the analysis of Criterion C.

Discussion

The primary purposes of Experiment 2 were to replicate the recognition test pattern of activation and inhibition from Experiment 1 and to determine if inducing intention before acquisition impacts the acquisition performance or the learning of the task.

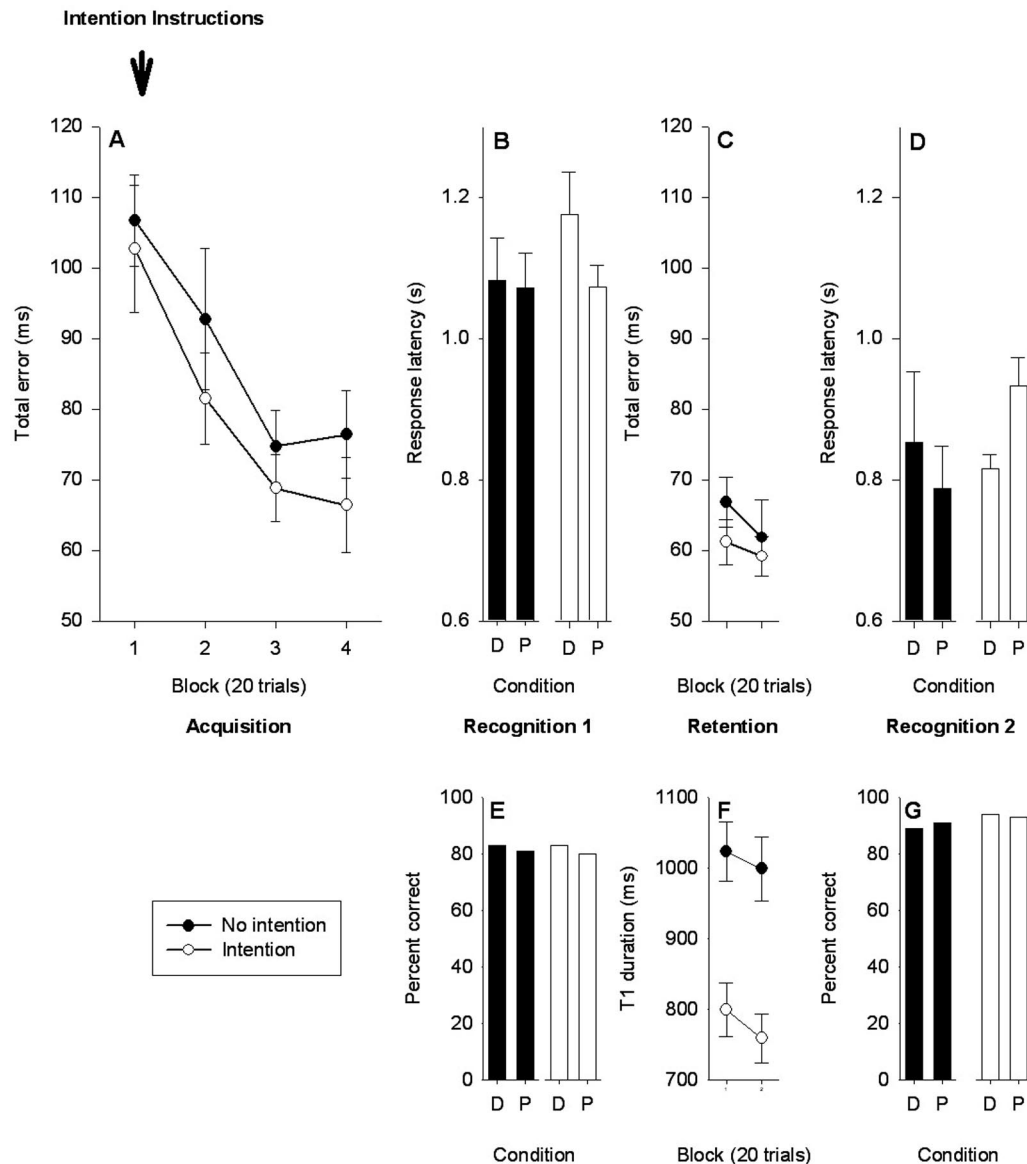


Figure 3. Mean total error during acquisition and retention for Experiment 2 are illustrated in Panels A and C, respectively. Mean response latency on Recognition Tests 1 and 2 are illustrated in Panels B and D, respectively (D indicates distractor patterns, and P indicates practiced patterns). Mean percentage correct on Recognition Tests 1 and 2 are illustrated in Panels E and G, respectively. Time required to initiate responses (T1) on the retention test is provided in Panel F. Intention instructions were provided to the intention group at the beginning of acquisition. Error bars represent standard errors.

The pattern of results on the recognition tests administered prior to and after the retention test was again consistent with the intention superiority effect found for verbal scripts and closely replicated the results from Experiment 1. As in Experiment 1, this effect was not confounded by differences in the percentage of correct responses to the practiced and distractor tasks on the recognition tests. Thus, it appears that the intention superiority effect results when intention instructions are provided prior to practice of the prospective tasks.

However, contrary to our preliminary hypothesis, providing the intention instructions prior to practice did not result in enhanced

performance either during acquisition or on the retention test. The results clearly demonstrated that prospective and distractor tasks were not differentially learned, even if intention is provided before encoding. Indeed, performance of the two groups remained remarkably similar regardless of whether intention was induced or not. Even the decreased time required to initiate the task on the retention test was similar to that observed in Experiment 1. Thus, on the basis of this experiment, we must again conclude that intention affects the activation and inhibition of the memories underpinning the tasks and time required to prepare a response for execution, but not response production.

Experiment 3

In Experiment 1, where intention instructions were provided after acquisition, and in Experiment 2, where instructions were provided prior to acquisition, intention played little or no role in the performance or learning of the tasks. The only influence on performance was on the time required to prepare the response. We had initially hypothesized that intention would enhance learning based on a recent observational learning experiment of ours (Badets et al., in press), in which intention to perform a motor skill in the future had a profound effect on the physical performance of the task. Clearly, intention to perform the task in the future caused observers to process task-related information differently than when they were simply attempting to understand the demands of the task, but the results of Experiments 1 and 2 suggest that this is not the case for physical practice.

The primary purposes of Experiment 3 were to replicate the findings from the recognition tests from Experiments 1 and 2 using a within-subjects design and to extend these findings to observational learning. By including a yoked observation group, we should be able to replicate our earlier findings relative to enhanced learning with intention compared with intention to produce the observed task and to determine if similar activation and inhibition patterns associated with intention occur for observational learning as have been observed for physical practice. Numerous experimental manipulations have been shown to impact observational learning in a manner very similar to that for physical practice. However, it is possible that other manipulations may result in differential findings. Determining the differences and similarities between physical and observational effects should provide important clues as to the processes involved in both modes of practice. Thus, based on the results of Experiments 1 and 2 in which intention before or after encoding failed to impact physical task performance, we do not expect intention instructions provided during acquisition to impact the learning of the task for the physical practice group. However, based on our previous experiment with intention in observational learning (Badets et al., in press), we do expect intention to positively affect the learning of the tasks through observation.

Method

Participants. University students ($N = 18$) volunteered to participate in the experiment (mean age 20.8 years, $SD = 1.4$ years). None of the participants had prior experience with the task or was informed about the purpose of the experiment. Each participant was requested to read and sign an informed consent form prior to participation in the experiment.

Apparatus and task. The apparatus and tasks were identical to that used in Experiments 1 and 2. However, two of the four tasks were identified as a "test task." This was accomplished by displaying the words *Test Task* above the goal movement pattern each time these tasks were displayed during acquisition. Participants were informed that they would be tested on the test tasks following acquisition. No text was displayed when the other two tasks were presented. In addition, a monitor was not used to display the goal pattern and feedback; rather, the display was projected on the wall in front of the performer. This larger display was used so that both the physical practice participant and an observer would have a clear view of the goal pattern and feedback.

Procedure. Two participants reported to the laboratory for each testing session. After completing the informed consent and reading the general instructions, participants were assigned to either a physical or observational

practice group by a coin toss. The physical practice participants were told they would physically practice four related tasks. The observational practice participants were told that they should learn the task through observation. Both physical and observational participants were informed that learning all the tasks was important, but that they would be tested on the test tasks in a retention test following acquisition. The long-left-1 and short-right-1 tasks were designated as test tasks and the short-right-1 and long-right-1 tasks were presented without designation. In all other respects, the acquisition and retention phases were conducted in a manner identical to that in Experiments 1 and 2.

At the end of the retention interval, both physical and observational participants performed a recognition test consisting of 20 trials of the test tasks, 20 trials of the nondesignated but practiced tasks, and 20 trials using unpracticed task variations. The three types of tasks were presented in a quasi-random order. These tasks were labeled prospective (P), distractor (D), and neutral (N), respectively. After this recognition test, all participants were asked to complete a retention test consisting of 40 trials. The retention test was conducted in a manner identical to that of acquisition except that the designation *Test Task* was not displayed for any of the tasks. After the retention phase, all participants performed the second recognition test in the same manner as the first.

Results

Acquisition and retention. Timing errors for the physical practice group during acquisition were analyzed in a 2-Task (distractor vs. prospective) $\times$ 4-Block (1–4) ANOVA with repeated measures on task and block (Figure 4A and C, respectively). The main effect of block was significant, $F(3, 48) = 7.16, p < .01$. Duncan's new multiple-range tests indicated that errors were higher in Block 1 than in Blocks 2–4, which did not differ from each other. The main effect of task, $F(1, 48) < 1, p > .05$, and the Task $\times$ Block interaction, $F(3, 48) < 1, p > .05$, were not significant.

Timing errors and T1 for the physical and observational groups during retention were analyzed in separate Task (prospective vs. distractor) $\times$ Block ANOVAs with repeated measures on both factors. Timing errors for the physical practice group on the retention test did not differ across the two types of tasks, $F(1, 24) = 1.19, p > .05$. However, the analysis on the retention performance for the observational group indicated improved performance on the prospective tasks relative to the distractor tasks, $F(1, 24) = 6.22, p < .01$. The main effects of block and the Task $\times$ Block interactions for both the physical and observational practice groups failed significance.

The analyses of T1 indicated a main effects of task for the physical practice, $F(1, 24) = 5.07, p < .05$, and observational practice, $F(1, 24) = 3.69, p < .05$, groups. Both groups initiated responses to the prospective task with shorter latencies than to the practiced but distractor tasks. The main effects of block and the Task $\times$ Block interactions for both groups failed significance.

Recognition tests. Recognition test reaction times and percentage correct for the physical and observational practice groups were analyzed in separate 3-Task (prospective, distractors, neutral) $\times$ 2-Test (preretention vs. postretention) ANOVAs with repeated measures on task and test (Figure 4B and D). The main effect of test for the observation group was significant, $F(1, 42) = 4.50, p < .05$, with the reaction times shorter on the postretention recognition test than on the preretention recognition test. The main effect of test for the physical practice group was not significant, $F(1, 42) < 1, p > .05$. While the main effect of task was not significant for either the physical or observational group, $F(1, 42) < 1, p > .05$,

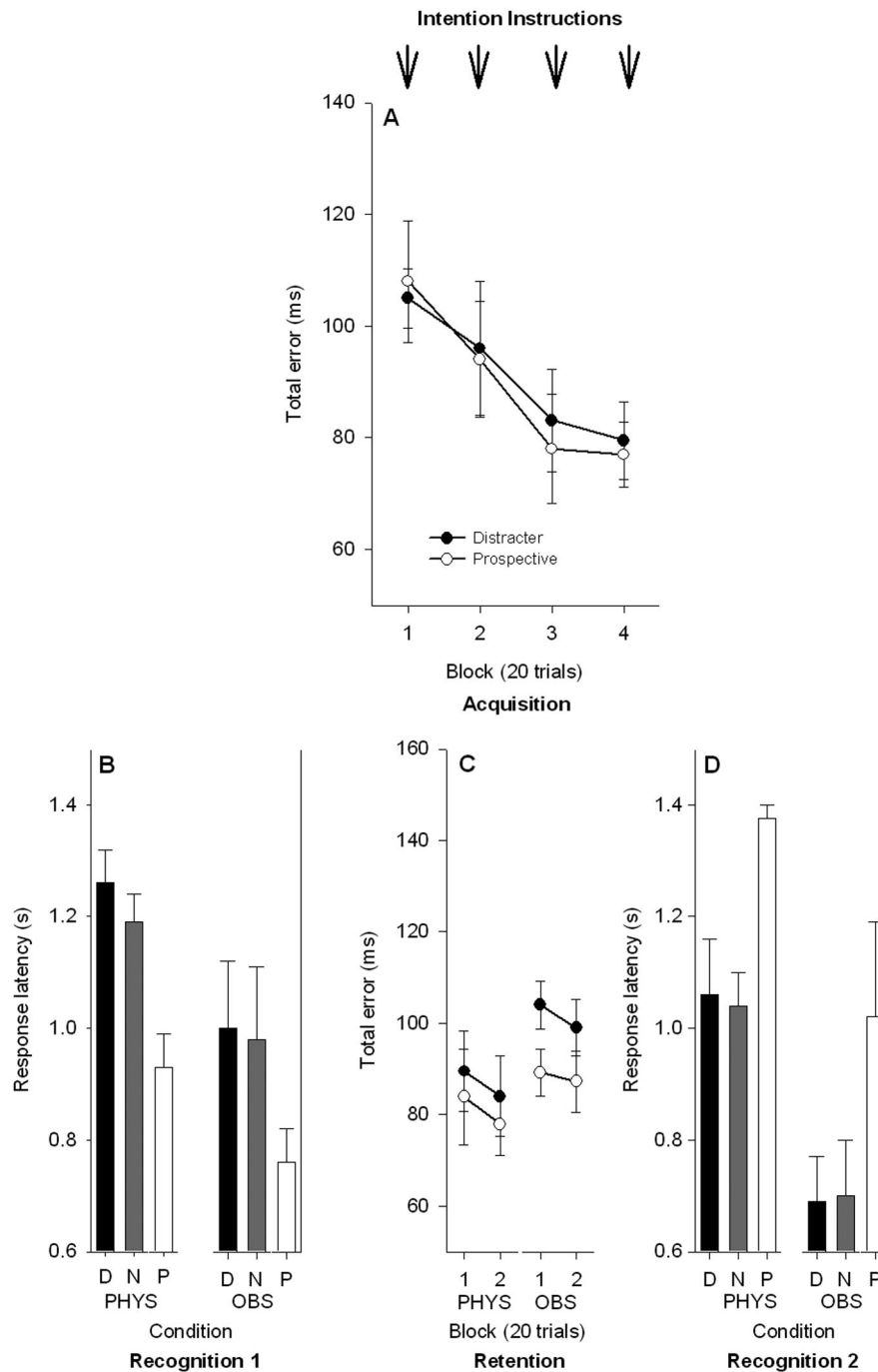


Figure 4. Mean total error during acquisition and retention for Experiment 3 are illustrated in Panels A and C, respectively. Mean response latency on Recognition Tests 1 and 2 are illustrated in Panels B and D, respectively. Intention instructions were provided to the intention group at the beginning of and during acquisition (D indicates distracter patterns that were practiced but without intention, N indicates patterns that were not practiced, and P indicates practiced patterns with intention). Error bars represent standard errors.

the Task $\times$ Test, $F(2, 42) = 8.28$, interaction for the physical practice group and for the observational group, $F(2, 42) = 11.72$, $p < .01$, were significant.

Simple main effect analysis on the preretention recognition test failed to detect differences between latencies to distractors and

neutral tasks for either the physical or observational groups, but did detect shorter reaction times for the prospective tasks for both groups. On the postretention recognition test, simple main effect analyses again failed to detect differences between distractors and neutral tasks for either the physical or observational group, but did

detect differences for the prospective tasks for both groups. However, on the postretention recognition test, latencies were longer for the prospective tasks than for the distractors or neutral tasks for both the physical practice and observational groups.

The analysis of percentage correct failed to detect any significant main or interactive effects for either the physical or observational practice groups.

Discussion

One purpose of Experiment 3 was to use a within-subjects design to replicate the results from Experiments 1 and 2, which used a between-subjects design. In the within-subjects design, participants practiced four tasks. They were informed that practice on all tasks was important but that they would be tested on two of the tasks. The test tasks were identified throughout acquisition. The pattern of results on the recognition tests for the physical practice participants, however, was remarkably similar to those from the earlier experiments. On the first recognition test, latencies were shorter for the two prospective tasks than on the two tasks that were practiced, but which they did not expect to perform on the delayed retention test. As in earlier experiments, acquisition and retention performance of the physical practice group was not affected by the intention instructions.

A second purpose was to determine if the pattern of latencies on the recognition tests and pattern of performance on the retention test would be similar for observational learners, who received the same intention instructions as the physical practice performers but did not physically perform the tasks until the retention test. Remarkably, the pattern of latencies on the recognition tests for the observers was very similar to that found for the physical practice participants. Thus, it seems the intention superiority effect holds for observational practice as well as for physical practice. This adds to a growing list of effects that act on physical and observational learning similarly. The performance pattern found on the retention test for observational participants, however, was different from that found for the physical practice participants. In Experiments 1–3, the performance of participants that physically practiced the tasks on the retention test did not differ as a result of intention instructions or when the intention instructions were provided (intention established after, before, and during acquisition, respectively). Intention instructions, however, resulted in enhanced performance on the prospective tasks for the observational practice group. These retention findings are consistent with Badets et al. (in press). Further, these findings of different patterns of results for observational and physical practice participants on the retention test but similar patterns on the recognition tests may suggest independent processes are responsible for these findings. This notion will be discussed in more detail in the general discussion.

General Discussion

In three experiments, we found consistent patterns of activation for prospective tasks prior to the retention test and inhibition of these tasks after the intention was fulfilled in the retention test. This pattern was remarkably similar for between-subjects (Experiments 1 and 2) and within-subjects designs (Experiment 3) and was demonstrated for both physical (Experiments 1–3) and observational practice (Experiment 3). These findings provide strong evidence for the intention superiority effect in motor skills. How-

ever, in physical practice we found no evidence of intention affecting the production of the task, regardless of whether the intention was established before, during, or after encoding. The only performance-associated effect of intention was a decreased time to retrieve and prepare the response for execution. This effect, however, may provide an important advantage for prospective tasks in many sport situations and may have implications for various lines of research that focus on the processing that occurs in the study time or reaction time intervals (e.g., Immink & Wright, 2001) and is consistent with the notion that unfulfilled intentions are stored in a heightened state of activation.

Intention Superiority Effect

The intention superiority effect refers to the notion that memories for intended actions are stored in a heightened state of activation or otherwise afforded a privileged status in memory until the intention is fulfilled or cancelled, after which it is inhibited. Presumably, the heightened state of activation results in the memory being more accessible than retrospective memories—effecting an increased chance of remembering the intention and, thus, fulfilling the intention. This heightened state of activation has been referred to as an increased *task tension* by Lewin (1926; also see Zeigarnik, 1927) and more recently by Anderson (1993). Zeigarnik, for example, found superior recall of activities interrupted prior to completion compared to the recall of activities that had been completed. In Anderson's adaptive control of thought (ACT*) model, he proposed goal nodes. These nodes were proposed to exhibit high and constant activation that is sustained in working memory without the need for rehearsal. Similarly, Goschke and Kuhl (1993) have proposed a persistence hypothesis, whereby increased activation is described in terms of the persistence of the memory trace (resistance to decay or interference). Indeed, a number of experiments have reported the persistence of intentions across a wide range of intervals ranging from a few minutes to several days (e.g., Harris & Wilkins, 1982; Wilkins & Baddeley, 1978).

Other theoretical perspectives (Anderson & Lebiere, 1998; Newell, 1990) describe intentions effects, not in terms of activation and inhibition, but in terms of *goal stacks* where intentions are ordered (stacked) by age with the newest intention on the top where it is most accessible. Older unfilled intentions are stacked below in reverse chronological order (new on top and oldest on the bottom). This notion has great intuitive appeal because this type of ordering of tasks is often used in computer software and computational cognitive models. Indeed, many of us keep lists of things to do whereby the most recent is on the top of the stack and the oldest on the bottom. This proposal is interesting because it implies rather straightforward predictions concerning the relationship between the chronology with which intentions are established and the activation levels of these intentions. Although the present experiments do not directly address this issue, one would expect that if two intentions are established, Intention A before Intention B, that Intention B would exhibit heightened activation until completion relative to Intention A. After fulfillment of Intention B, Intention A should exhibit a spontaneous increase in activation levels. The finding that fulfilled intentions are associated with decreased activation is consistent with this notion. Indeed, the present paradigm could be modified in order to establish two intentions.

It is important to note that the present data demonstrate a degree of dissociation between the activation of memory underpinning the intention and the content of that memory. That is, heightened activation of a memory resulting from forming an intention to use the memory in the impending future does not seem to affect the content of the memory and, thus, does not affect the production of the prospective task. Remarkably, intention instructions did not affect the learning of the movement sequences in the present experiments regardless of whether the intention was established before, during, or after acquisition. The prospective tasks were performed with no better precision than tasks for which no intention was established. Consistent with the latency data from the first recognition test, where prospective tasks were recognized more quickly than distractor tasks, participants initiated their responses more quickly for prospective tasks than for distractor tasks. These findings of dissociation are consistent with earlier assumptions and findings that memory for intent was dissociated from the content of the memory (e.g., Einstein & McDaniel, 1990; Reason, 1984). Indeed, Kvavilashvili (1992) concluded after reviewing experiments on intention that remembering an intention is independent of remembering how to do the intended task.

One persistent question, however, is whether or not specialized memory systems are required to account for the intention superiority effect. Recently, Altmann and Trafton (2002) concluded that special memory structures which exhibit persistent task tension or maintain heightened activation levels independent of rehearsal, associate priming, or strategic coding are not necessary to account for the effect. Rather, they argue that small windows of opportunity in the designs may allow rehearsal, associative priming, and strategic encoding that could account for the effect without implicating some specialized memory depository. In the present designs, however, there were multiple opportunities for additional processing, especially in Experiments 2 and 3 in which intention instructions were provided before or during the encoding process. Nevertheless, there was virtually no difference in the recognition results when the instructions were provided at the end of practice. We would have expected that, if additional processing of the prospective tasks were undertaken, performance advantages would be observed on not only the recognition test but also the retention test. This was not the case. Prospective tasks, where intention was established, were performed no better than distractor tasks, where no intention instructions were provided. While it is possible that additional processing had selective effects only on the activation of the memory and not the content of the memory, this seems unlikely. Moreover, an explanation for the pattern of activation on Recognition Test 1 based solely on selective encoding does not seem consistent with the pattern of inhibition observed on Recognition Test 2. Selective encoding of the prospective tasks should lead to similar performance on both recognition tests. Indeed, selective encoding would also be expected to show up in enhanced acquisition and retention performance. This clearly was not the case.

Intention Superiority Effect and the Subject-Performed Task Effect

An alternative explanation for the intention superiority effect observed in earlier experiments (e.g., Goschke & Kuhl, 1993; Marsh et al., 1998, 1999) is based on the notion that motor and

sensory information may be spontaneously activated for verbal materials that are intended for future execution (Freeman & Ellis, 2003). Thus, a verbal script like "setting the table" or even a single verb like "clap" may activate motor and sensory information associated with the action of setting the table or clapping one's hands whereby the additional sensory-motor information and associated processing strengthens accessibility and retention performance. Tettamanti et al. (2005) using fMRI reported that even listening to action-related sentences activated the premotor cortex where actions are motorically encoded.

More specifically, Freeman and Ellis (2003) argue that, if the intention superiority effect is a result of the privileged status of prospective memories, this effect should be observed irrespective of whether the prospective tasks are encoded verbally or through enactment (motorically). However, if the pattern of activation observed for prospective versus neutral tasks in earlier experiments is a result of additional sensory or motor processing of the prospective tasks, the effect should be reduced or eliminated when all the tasks are enacted at encoding. In four experiments when the words (verbs) were enacted during encoding, the analyses failed to detect significant intention effects (reduced latencies on the prospective task relative to distractor and neutral tasks) on a recognition test administered before the recall test (no postrecall recognition test was administered). Freeman and Ellis maintain that these findings were consistent with an action-superiority account for the intention superiority effect and not consistent with explanations based on privileged status.

According to this account, intention to perform a set of tasks in the impending future should have little or no effect when all tasks (prospective and neutral) are enacted during encoding. Although they did not assess postrecall activation, Freeman and Ellis (2003) argue that the action-superiority account does not predict the inhibition of the action representation following fulfillment of the intention on the retention test. That is, additional processing of sensory and motor information related to the action leads to enhanced accessibility of the memory for the action regardless of whether the recognition test is administered before or after the fulfillment of an intention to perform. Clearly, the findings from the present experiments are not consistent with either prediction. In the present experiments, all participants enacted (physically practiced) the tasks during acquisition. The only difference between groups in Experiments 1 and 2 was the intention instructions. In Experiment 3, the only difference between tasks was whether or not specific tasks were designated for later recall. However, shorter latencies were observed for the intention group on Recognition Test 1, which was administered prior to fulfillment of the intention, than for the group not provided intention instructions. This pattern of results was observed irrespective of whether the intention instructions were provided before (Experiment 2), during (Experiment 3), or after encoding (Experiment 1). More important, the action-superiority account, as well as other accounts based on additional or more effective encoding of the intentional tasks, does not predict the inhibition observed on Recognition Test 2 in all three experiments following fulfillment of the intention. In the three present experiments, inhibition of the prospective tasks was observed relative to neutral and distractor tasks when the intention was fulfilled.

Obviously, it will be important in future experiments to sort out the reason or reasons for these discrepant findings, although a few

clues may currently exist. In one case, the translation of verbal codes into action essentially eliminates the pattern of activation typically observed for the prospective tasks. In the present experiments, translation of a visual code (picture of the timing pattern) into action resulted in a pattern of activation and inhibition consistent with the intention superiority effect. It is possible that different types of sequential representations underpin verbal scripts that are enacted and motor sequences with the type of processing relying on different neural pathways (see Keele, Ivry, Mayr, Hazeltine, & Heuer, 2003, for a review). It is interesting that the timing tasks used in the present experiments do not seem to lend themselves readily to verbal description, although a verbal description of the tasks is possible. We have used these tasks before, and when participants were asked to describe the tasks in postexperiment interviews, the participants most often described the sequences by tapping the timing sequence out with their fingers or by repeating a simple sound (like "da") with specific temporal intervals between the taps or sounds. It seems like verbal descriptions of the timing tasks, while possible, are not directly related to the way in which participants code the sequences. Thus, the type of representation (verbal or nonverbal) may determine the magnitude of the intention effect in the presence of enactment effects. In addition, it is also possible when verbal scripts or verbs are enacted that both verbal and nonverbal codes are established. These codes may result in a form of competition or interference that delays responding (see Jahn & Engelkamp, 2003, for a discussion). The timing tasks used in the present experiment are less likely to result in a competition or interference between codes because verbal codes, if any, do not appear to be developed to the same extent as the nonverbal codes.

Other distinctions between experiments involve previous knowledge of the task and the amount of practice. In the Freeman and Ellis (2003) experiments, common verbs were enacted (the verb *clap* was enacted by clapping). Thus, the participant may have augmented the verbal code through enactment. In the present experiment, a series of keypresses was learned during the acquisition period. The acquisition data clearly illustrate the relatively poor performance of the participants early in practice relative to late in practice and on the retention test. It is likely that a motor code for the task was developed that did not rely on or build on a previous verbal representation for the task. In addition, in the Freeman and Ellis experiments each verb was read aloud or enacted only once in the initial phase of the experiments. In the present experiments, the tasks were practiced in four blocks of 20 trials. Thus, the effect may be related to the code established for the task, the amount of past experience with the task, or both. The bottom line is that strong evidence for the intention superiority effect was found for timing tasks even though all tasks (prospective and distractor) were enacted during acquisition.

Effects of Intention on Observational Learning

In a recent paper (Badets et al., in press), we found that observers who were informed that they would be required to produce a specific movement sequence physically in the future (intention) performed the task on their initial physical practice trials substantially better than a group of participants who were told that they will be questioned about the details of the response at a later time

but not told they would be required to perform the task physically. This result was replicated and expanded on in Experiment 3 in this paper. In Experiment 3, observers, informed that they would be required to produce a set of two responses in the impending future, performed these responses significantly better than another set of responses that they also practiced but for which no intention instruction was provided. In contrast, intention instructions provided to physical practice participants in all three experiments failed to result in any performance increment. However, observers exhibited the same pattern of activation prior to the retention test and inhibition after the retention test, as found for the physical practice participants on the prospective tasks. Thus, intention appeared to affect activation and inhibition patterns for observers and physical practice participants similarly, but only the performance of the observers benefited from intention.

Recent neuroimaging experiments report that a set of common neural structures are activated during both action production and action observation (Gallese & Goldman, 1998; Grèzes & Decety, 2001; see Jeannerod, 1999, for reviews) only when the observers intend to perform the task in the future. The shared neural structures included the premotor cortex, supplementary motor area, inferior parietal lobule, cingulate gyrus, and cerebellum. Using positron-emission tomographic measures, Decety et al. (1997) reported that observation with the intent to reproduce the observed action was associated with cerebral activation in the regions involved in the planning and in the generation of actions (see also Grèzes, Costes, & Decety, 1999). These findings led Decety and Grèzes (1999) to propose that the neural substrate for action planning is activated during perception of action only when the intention is to imitate that action at some time in the future. Consequently, at a behavioral level, instructions that result in intention to physically perform the task in the future should be beneficial when compared to no specific instruction. The working hypothesis is that the intention to perform a task physically later results in the activation of cognitive and motor neural structures, in the absence of overt action that facilitates learning of the task.

A major unanswered question, however, is whether or not the relationship between the effects of intention on observational learning and the effects of intention on patterns of activation arise from the same or different processes. More specifically, are the beneficial effects of intention during observational learning on later physical performance and the activation and inhibition patterns that are indicative of the intention superiority effect manifestations of one process or are they two independent processes that are coincidentally affected by intention? While the present data cannot answer this question directly, it does appear that these are independent processes. We base this preliminary conclusion on the findings that observation and physical practice both exhibit patterns of activation and inhibition characteristic of the intention superiority effect, but only observation learning benefits from intention instructions. If these effects were supported by the same or similar processes, we would expect similar patterns not only for activation patterns but also for performance. However, it should be noted that the performance of the observation group was somewhat poorer than that of the physical practice group on the retention test. Thus, the observation group had more room to improve than the physical practice group.

Conclusion

Intention to perform a motor sequence in the near future results in increased levels of activation reflected in reduced latencies to recognize the sequence pattern and in reduced latencies to retrieve, program, compile, or otherwise ready the task for execution. After the intention is fulfilled by the execution of the prospective task in the retention test, the recognition latencies are inhibited relative to tasks that were neutral with respect to intention. In addition, intention resulted in enhanced observational learning, but had little or no effect on physical practice learning. This latter finding may prove to be interesting because further study comparing the effect of intention on physical and observational learning may provide clues as to whether or not the activation and inhibition patterns produced in observation and physical practice arise from the same mechanism as the enhanced observational learning. Our preliminary hypothesis is that these are independent mechanisms or processing effects; however, this is based solely on the different pattern of effects relative to observational and physical practice learning but similar patterns for activation and inhibition.

Additional work is needed to understand better the relationships between retrospective and prospective memories in terms of the retrieval, execution, and learning of motor skills. To our knowledge, the present experiments are the first that contrast the learning of prospective and retrospective motor tasks. While the results demonstrate that this distinction does not impact the production or learning of the motor skills per se, it does suggest that the time required to retrieve, program, or otherwise ready the memories underpinning the response for execution are affected by whether or not an intention has been established. The reduced latencies associated with prospective tasks but not tasks neutral with respect to intention, although relatively small (e.g., 200 ms) could be the difference between success and failure in many real world tasks. Indeed, the savings of 200 ms for a military pilot, for example, who intends to execute an evasive maneuver when a certain situation (event based) or time (time based) arrives, may be the difference between life and death. In any event, the full impact of intention on the control and learning of motor skills deserves additional study.

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